



The Characteristic Properties of Acids & Bases

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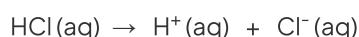
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- * Proton Transfer, Strong & Weak Acids
- * Classifying Oxides



Properties of acids

- Acids have pH values of below 7, have a sour taste (when edible) and are corrosive
- Acids are substances that can neutralise a base, forming a salt and water
- When **acids** are added to water, they form positively charged **hydrogen ions** (H^+)
- The presence of H^+ ions is what makes a solution acidic

Example: Hydrochloric acid



Acids and metals

- Only metals **above hydrogen** in the reactivity series will react with dilute acids.
- When acids react with metals they form a salt and hydrogen gas:



- The name of the salt is related to the name of the acid used, as it depends on the **anion** within the acid

Examples of the names of salts from specific acids and metals

Acid	Name of products	Equation for reaction
Hydrochloric acid	Magnesium chloride and hydrogen	$Mg + 2HCl \rightarrow MgCl_2 + H_2$
Sulfuric acid	Magnesium sulfate and hydrogen	$Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$
Nitric acid	Magnesium nitrate and hydrogen	$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$

Acids with bases

- Metal **oxides** and metal **hydroxides (alkalis)** can act as **bases**
- When they react with acid, a **neutralisation** reaction occurs
- In all acid-base neutralisation reactions, **salt** and **water** are produced



Examples of reactions between acids and bases

Acid	Name of products	Equation for reaction
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Hydrochloric acid	Magnesium chloride and water	$\text{Mg(OH)}_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$
Sulfuric acid	Magnesium sulfate and water	$\text{MgO} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2\text{O}$
Nitric acid	Magnesium nitrate and water	$\text{Mg(OH)}_2 + 2\text{HNO}_3 \rightarrow \text{Mg(NO}_3)_2 + 2\text{H}_2\text{O}$



Your notes

Acids with metal carbonates

- Acids will react with metal carbonates to form the corresponding metal salt, carbon dioxide and water:



Examples of reactions between acids and carbonates

Acid	Name of products	Equation for reaction
Hydrochloric acid	Magnesium chloride, carbon dioxide and water	$\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$
Sulfuric acid	Magnesium sulfate, carbon dioxide and water	$\text{MgCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$
Nitric acid	Magnesium nitrate, carbon dioxide and water	$\text{MgCO}_3 + 2\text{HNO}_3 \rightarrow \text{Mg(NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$

What are indicators?

- Two colour indicators are used to distinguish between acids and alkalis
- Many plants contain substances that can act as indicators and the most common one is **litmus** which is extracted from lichens
- Synthetic indicators are organic compounds that are sensitive to changes in acidity and appear different colours in acids and alkalis
- Thymolphthalein** and **methyl orange** are synthetic indicators frequently used in acid-alkali titrations

Two Colour Indicators Table

Indicator	Colour in acid	Colour in alkali
Litmus	red	blue

Thymolphthalein	colourless	blue
Methyl orange	red	yellow



Your notes

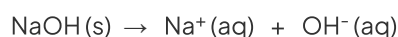
What are synthetic indicators?

- Synthetic indicators are used to show the endpoint in titrations as they have a very sharp change of colour when an acid has been neutralised by alkali and vice-versa
- Litmus is not suitable for titrations as the colour change is not sharp and it goes through a purple transition colour in neutral solutions making it difficult to determine an endpoint
- Litmus is very useful as an indicator paper and comes in red and blue versions, for dipping into solutions or testing gases

Properties of bases & alkalis

- Bases have pH values of above 7
- A base which is water-soluble is referred to as an **alkali**
- In basic (alkaline) conditions red litmus paper turns **blue**, methyl orange indicator turns **yellow** and thymolphthalein indicator turns **blue**
- Bases are substances which can neutralise an acid, forming a salt and water
- Bases are usually **oxides** or **hydroxides** of metals
- When **alkalis** are added to water, they form negative hydroxide ions (OH^-)
- The presence of the OH^- ions is what makes the aqueous solution an alkali

Example: Sodium hydroxide



Bases and acids

- When bases react with an acid, a neutralisation reaction occurs
- Acids and bases react together in a neutralisation reaction and produce a salt and water:



Examples of reaction between bases and acids

Acid	Name of products	Equation for reaction
Hydrochloric acid	Magnesium chloride and water	$\text{Mg(OH)}_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$
Sulfuric acid	Magnesium sulfate and water	$\text{MgO} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2\text{O}$

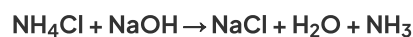
Nitric acid	Magnesium nitrate and water	$\text{Mg(OH)}_2 + 2\text{HNO}_3 \rightarrow \text{Mg(NO}_3)_2 + 2\text{H}_2\text{O}$
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Your notes

Alkalis and ammonium salts

- Ammonium salts undergo **decomposition** when warmed with an alkali
- Even though ammonia is itself a weak base, it is very **volatile** and can easily be displaced from the salt by another alkali
- A salt, water and ammonia are produced
 - For example:



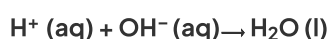
- This reaction is used as a chemical test to confirm the presence of the ammonium ion (NH_4^+)
- Alkali is added to the substance with gentle warming followed by the test for ammonia gas using damp red litmus paper
- The damp litmus paper will turn from **red** to **blue** if ammonia is present



Neutralisation reactions

What is a neutralisation reaction?

- Acids are a source of hydrogen ions, H^+
- Bases (or alkalis) are a sources of hydroxide ions, OH^-
- When they react together in a neutralisation reaction, the H^+ ions react with the OH^- ions to produce **water**
- This is the **net ionic equation** of all acid-base neutralisations and is what leads to a neutral solution, since water has a pH of 7:



- Not all reactions of acids are neutralisations:
 - For example when a metal reacts with an acid, although a salt is produced there is no water formed so it does not fit the definition of neutralisation



Examiner Tips and Tricks

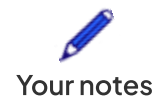
Not all reactions of acids are neutralisations. For example, when a metal reacts with an acid, although a salt is produced there is no water formed so it does not fit the definition of neutralisation.

Hydrogen ion concentration & pH

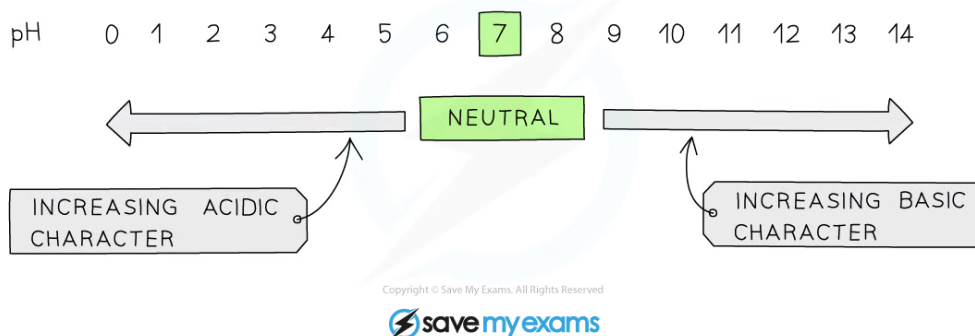
What is the pH scale?

- The pH scale goes from 1 – 14
- All acids have pH values of **below** 7, all alkalis have pH values of **above** 7
- The **lower** the pH then the **more acidic** the solution is:
 - pH 0–2 = typically strong acids (e.g. hydrochloric acid, sulfuric acid)
 - Extremely acidic substances can have values of below 1
 - pH 3–6 = typically weak acids (e.g. ethanoic acid, citric acid)
- The **higher** the pH then the **more alkaline** the solution is:
 - pH 8–11 = typically weak alkalis (e.g. ammonia solution, sodium hydrogencarbonate)
 - pH 12–14 = typically strong alkalis (e.g. sodium hydroxide, potassium hydroxide)

- A solution of pH 7 is described as being **neutral**



The pH scale



The pH scale showing acidity, neutrality and alkalinity



Examiner Tips and Tricks

The terms strong and weak refer to the degree of dissociation of the acid or alkali in solution, not to the pH alone.

A dilute strong acid can have the same pH as a concentrated weak acid.

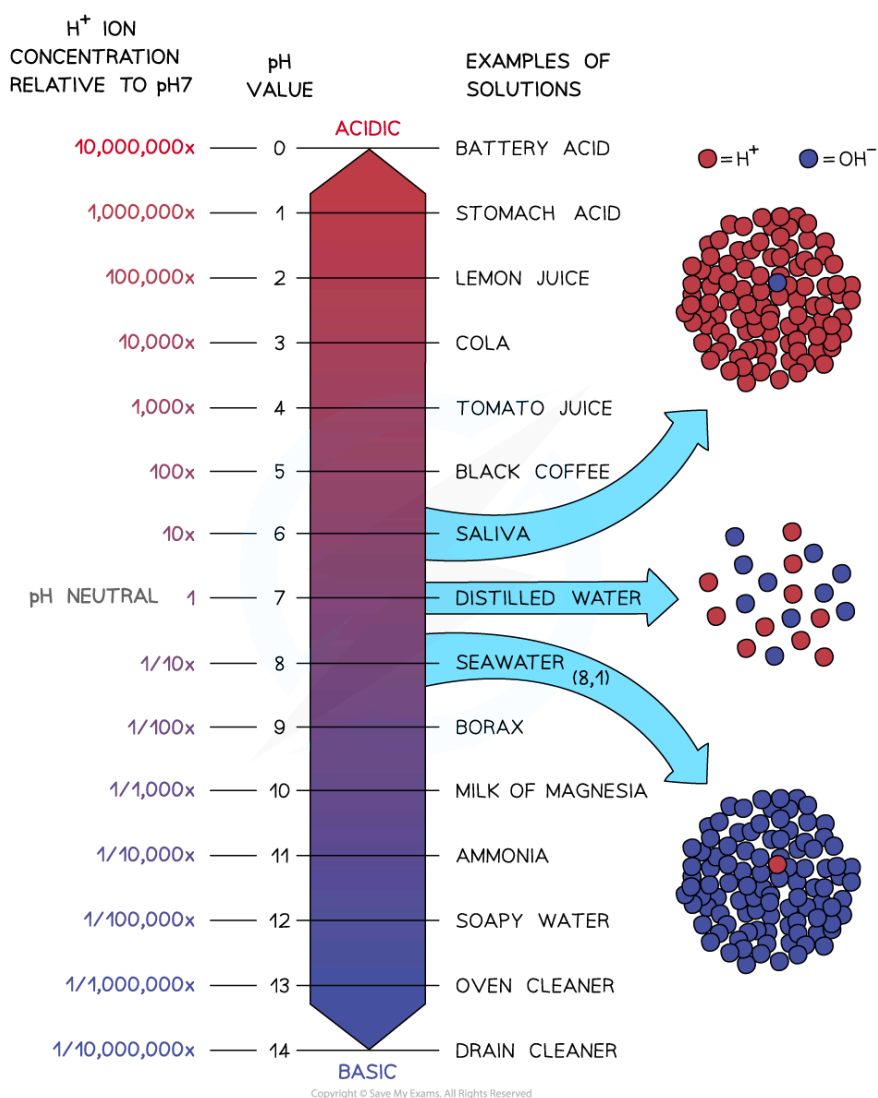
The pH scale and hydrogen ions

- We have already seen that acids are substances that contain hydrogen ions in solution
- The more hydrogen ions the stronger the acid, but the lower the pH
- The higher the concentration of hydroxide ions in a solution the higher the pH
- So pH is a measure of the concentration of H^+ ions in solution, but they have an inverse relationship
- The pH scale is **logarithmic**, meaning that each change of 1 on the scale represents a change in concentration by a **factor** of 10

Concentration of hydrogen ions and pH



Your notes



An acid with a pH of 3 has ten times the concentration of H⁺ ions than an acid of pH 4. An acid with a pH of 2 has 10 x 10 = 100 times the concentration of H⁺ ions than an acid with a pH of 4



Examiner Tips and Tricks

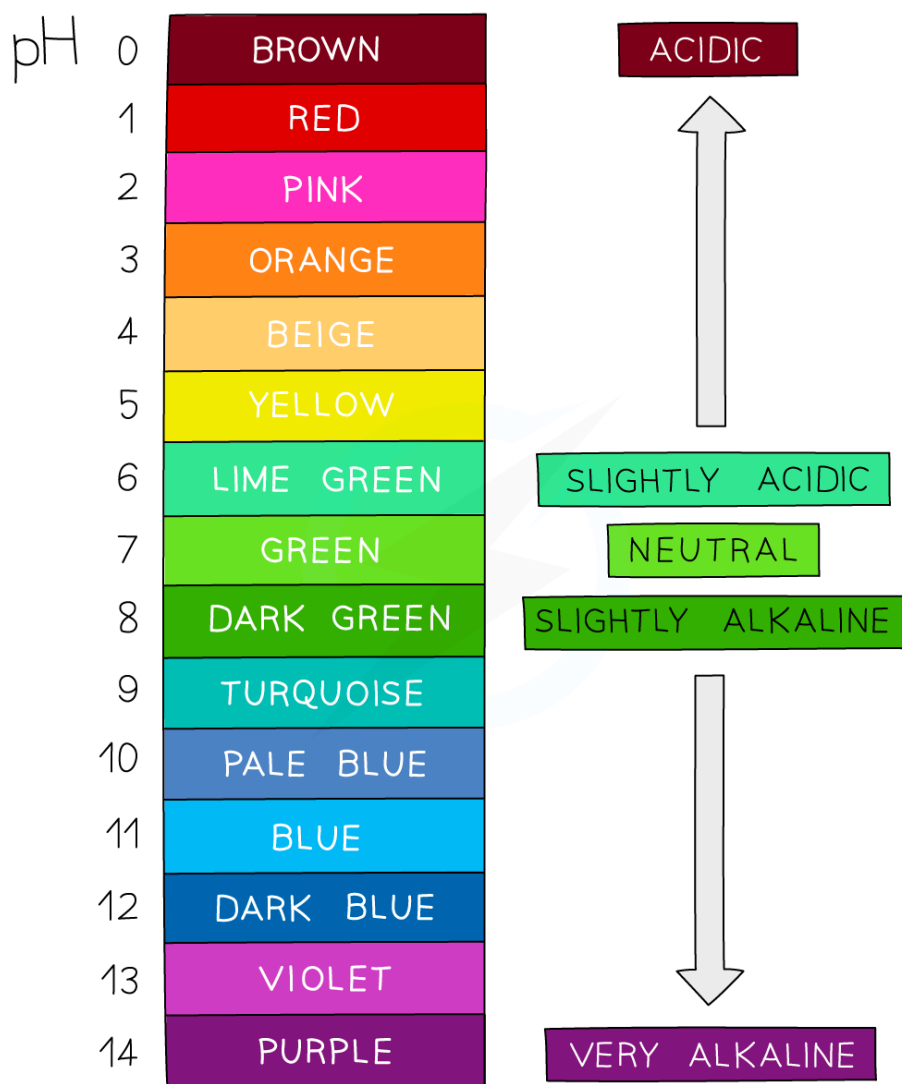
Acid strength is reflected in how many hydrogen ions are in solution. The more hydrogen ions the lower the pH and vice-versa.

How is universal indicator used?

- Universal indicator is a mixture of different **indicators** which is used to measure the pH
- A drop is added to the solution and the colour is matched with a colour chart which indicates the pH which matches specific colours



Your notes



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The pH scale with the Universal Indicator colours which can be used to determine the pH of a solution



Proton transfer, strong & weak acids

Proton transfer

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- Acids are **proton donors** as they ionise in solution producing protons, which are H^+ ions
- These H^+ ions make the aqueous solution acidic
- Bases are **proton acceptors** as they accept the protons which are donated by the acid

Proton transfer between hydrochloric acid and water

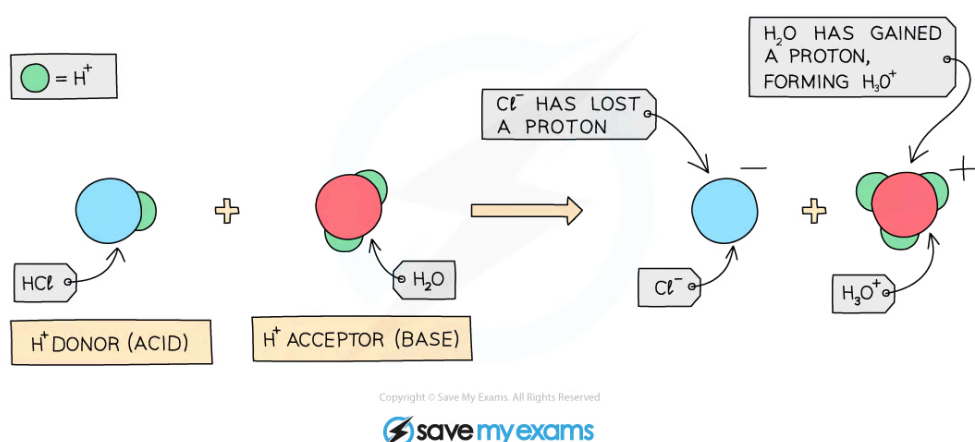
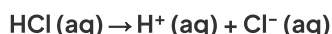


Diagram showing the role of acids and bases in the transfer of protons – here water acts as a base as it accepts a proton

What is a strong acid?

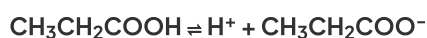
- Acids can be either **strong** or **weak**, depending on how many H^+ ions they produce when dissolved in water
- Strong acids **completely dissociate** (or ionise) in water, producing solutions of a **very low pH**
- Strong acids include HCl and H_2SO_4
- Example of a strong acid: hydrochloric acid



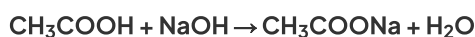
What is a weak acid?



- Weak acids **partially dissociate** (or ionise) in water and produce pH values which are closer to the **middle** of the pH scale, whilst still being below 7
- For weak acids, there is usually an **equilibrium** set-up between the molecules and their ions once they have been added to water
- Example of a weak acid: propanoic acid



- The equilibrium lies to the **left**, indicating a high concentration of intact acid molecules, with a low concentration of H^+ ions in the solution
- Another example of a weak acid is **ethanoic acid** which will react with alkalis such as sodium hydroxide to form ethanoate salts



Effect of concentration on strong and weak acids

- A concentrated solution of an acid is one that contains a **higher number** of acid **molecules** per dm^3 of solution
- It does not necessarily mean that the acid is strong
 - A concentrated weak acid only partly **dissociates** so the H^+ concentration remains relatively low

Comparing strong and weak acids at the same concentration:

- A strong acid at a given concentration has a lower pH than a weak acid at the same concentration
- This is because the strong acid completely dissociates, producing a higher concentration of H^+ ions
- A weak acid only partially dissociates, so fewer H^+ ions are produced and the pH is higher (less acidic)
 - For example, at the same concentration, hydrochloric acid has a lower pH than ethanoic acid
- Weak acids typically give pH values in the range 3 – 6, not pH 1 or 2

Reactivity:

- Since strong acids produce more H^+ ions at the same concentration, they are more reactive than weak acids
 - For example, hydrochloric acid reacts more vigorously with a metal or carbonate than ethanoic acid at the same concentration

Concentration vs strength:

- A dilute solution of a strong acid (e.g. HCl) can be more acidic than a concentrated solution of a weak acid (e.g. CH_3COOH), because the strong acid dissociates

completely even when dilute



Your notes



Classifying oxides

What are oxides?

- Oxides are compounds made from one or more atoms of oxygen combined with one other element
- Examples of oxides include:
 - Magnesium oxide - MgO
 - Zinc oxide - ZnO
 - Potassium oxide - K_2O
 - Carbon dioxide - CO_2
 - Sulfur dioxide - SO_2
 - Water - H_2O
- Oxides can be classified based on their acid-base characteristics

Acidic oxides

- Acidic oxides are formed when a **non-metal** element combines with oxygen
- They react with bases to form a salt and water
- When dissolved in water they produce an **acidic** solution with a **low** pH
- Common examples include CO_2 , SO_2 , NO_2 and SiO_2

Basic oxides

- Basic oxides are formed when a **metal** element combines with oxygen
- They react with acids to form a salt and water
- When dissolved in water they produce a **basic** solution with a **high** pH
- Common examples include CuO and CaO

Basic and acidic oxides



Your notes

BASIC OXIDES											ACIDIC OXIDES						
1/I 2/II		H										3/III 4/IV 5/V 6/VI 7/VII		0/VIII			
Li	Be											B	C	N	O	F	He
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac															
		Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu		
		Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr		

KEY

METAL

METALLOID

NONMETAL

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Metals form basic oxides while non-metals form acidic oxides

Amphoteric oxides

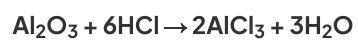
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Neutral oxides

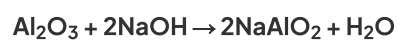
- Some oxides do not react with either acids or bases and thus are said to be **neutral**
- Examples include N_2O , NO and CO

Amphoteric oxides

- Amphoteric oxides are a curious group of oxides that can behave as **both acidic and basic**, depending on whether the other reactant is an acid or a base
- In both cases salt and water are formed
- Two of the most common amphoteric oxides are **zinc oxide, ZnO** and **aluminium oxide, Al_2O_3**
- The **hydroxides** of both of these elements also behave amphotERICALLY
- Example of aluminium oxide behaving as a base:



- Example of aluminium oxide behaving as an acid:



Your notes